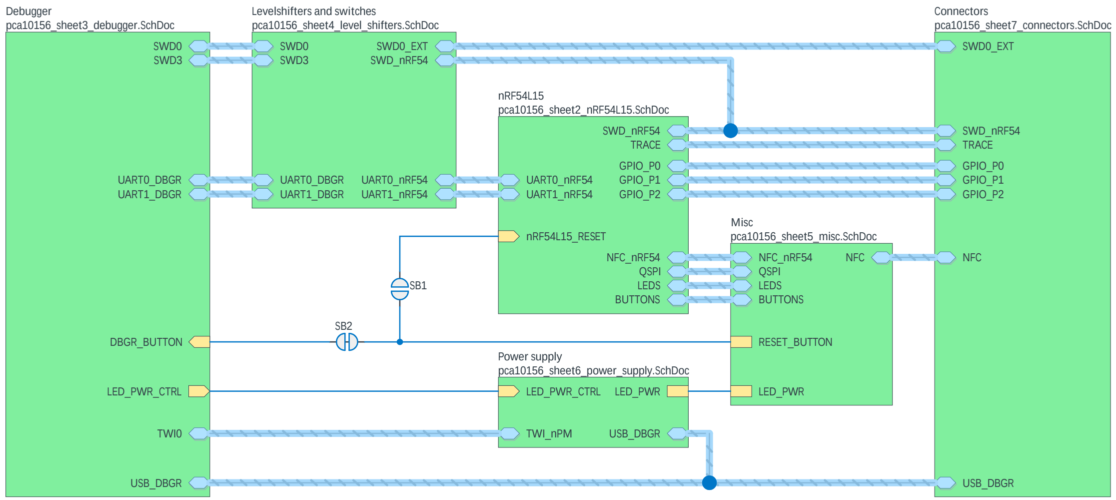
Nordic Semiconductor ASA

nRF54L15 Development Kit (PCA10156)



× The No ERC object is a design directive.
This directive is placed on a node in the circuit to suppress harmless warnings and/or erro r violation conditions that are detected when the schematic project is compiled.

Board fiducials

Logo's and markings

PCA10156
0.9.2
<year>.<week>
<UID>

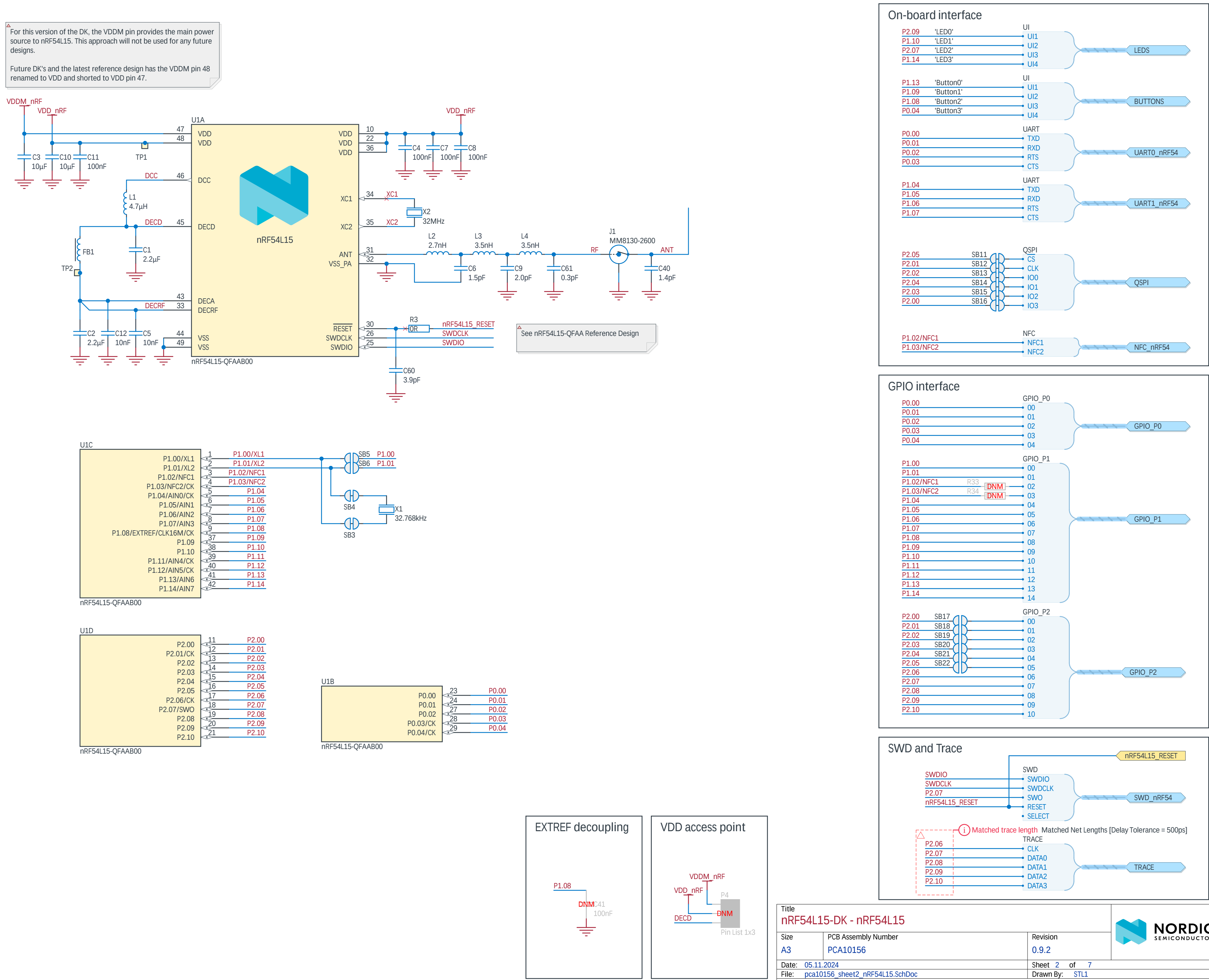
Segger Logo S

Nordic Logo

WEE Symbol

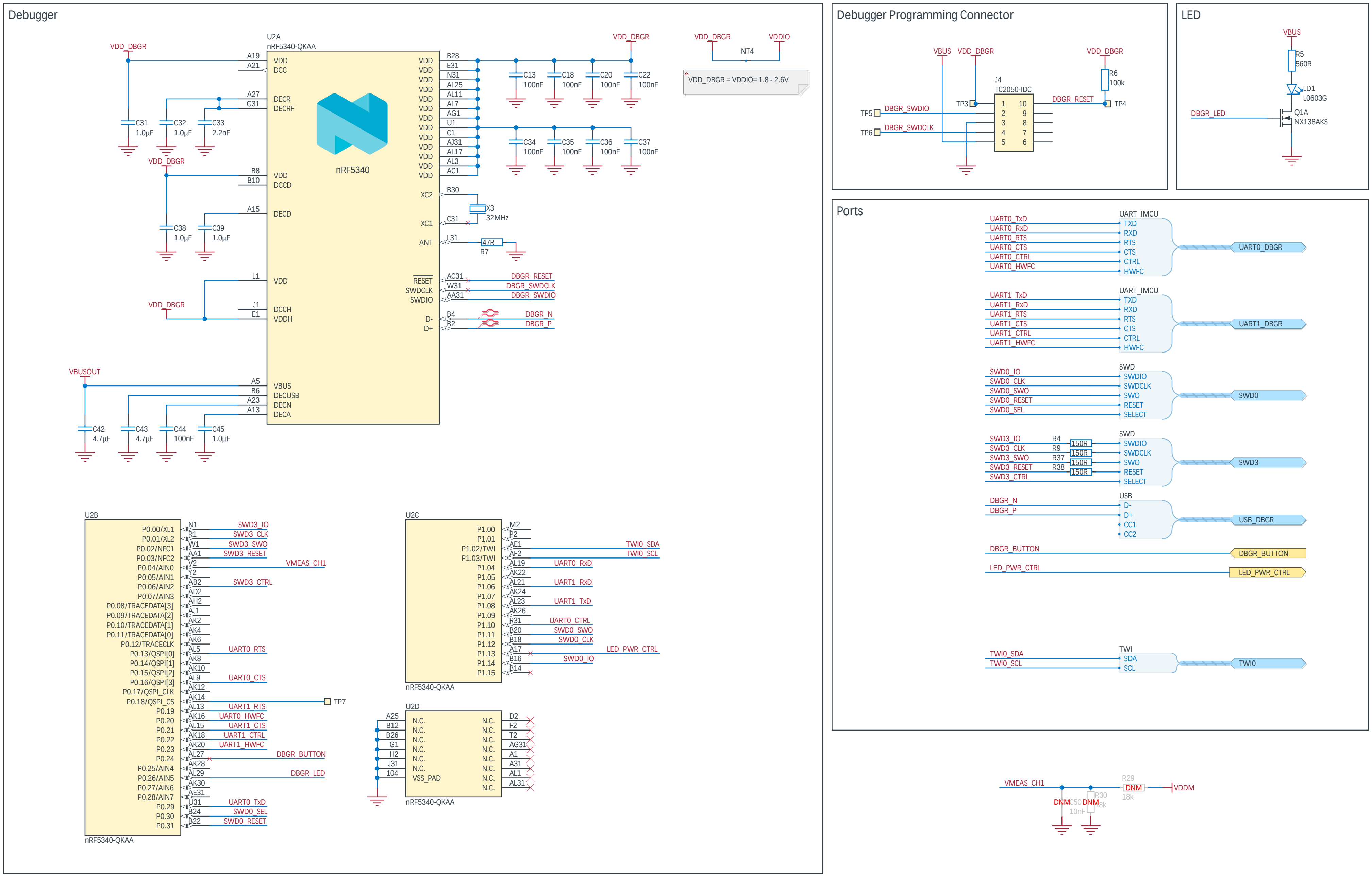
Title: nRF54L15-DK - Connections		
Size: A3	PCB Assembly Number: PCA10156	Revision: 0.9.2
Date: 02.10.2024		Sheet 1 of 7
File: pca10156_sheet1_connections.SchDoc		Drawn By: STL1
Classification: PUBLIC		

NORDIC SEMICONDUCTOR



Title: nRF54L15-DK - nRF54L15		
Size: A3	PCB Assembly Number: PCA10156	Revision: 0.9.2
Date: 05.11.2024		Sheet 2 of 7
File: pca10156_sheet2_nRF54L15.SchDoc		Drawn By: STL1
Classification: PUBLIC		

NORDIC SEMICONDUCTOR



Title: nRF54L15-DK - Debugger			
Size: A3	PCB Assembly Number: PCA10156	Revision: 0.9.2	
Date: 02.10.2024		Sheet 3 of 7	
File: pca10156_sheet3_debugger.SchDoc		Drawn By: STL1	
Classification: PUBLIC			

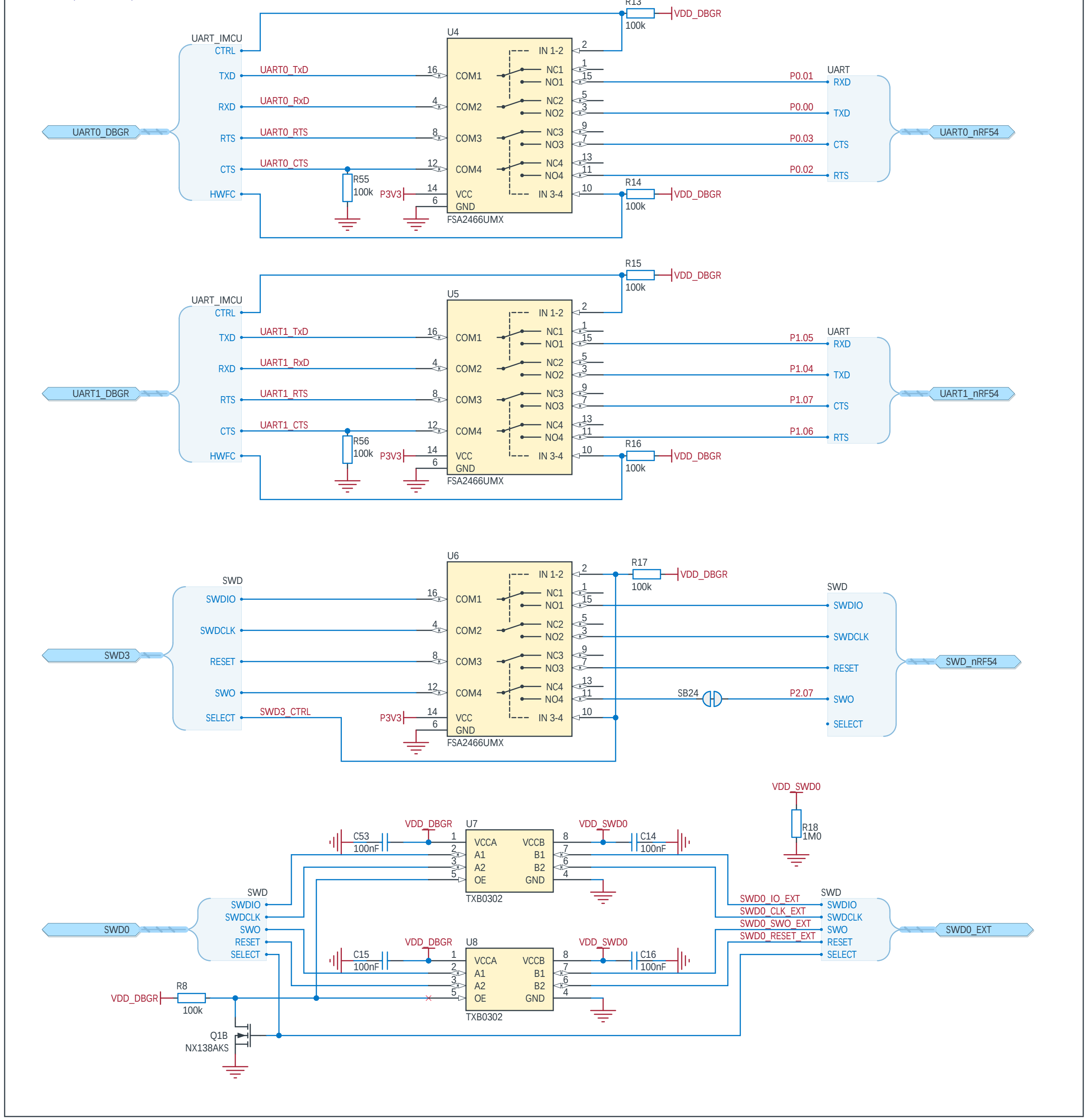
NORDIC SEMICONDUCTOR

FSA2466UMX:
NC/NO terminal: 6pF, OFF
21pF, ON

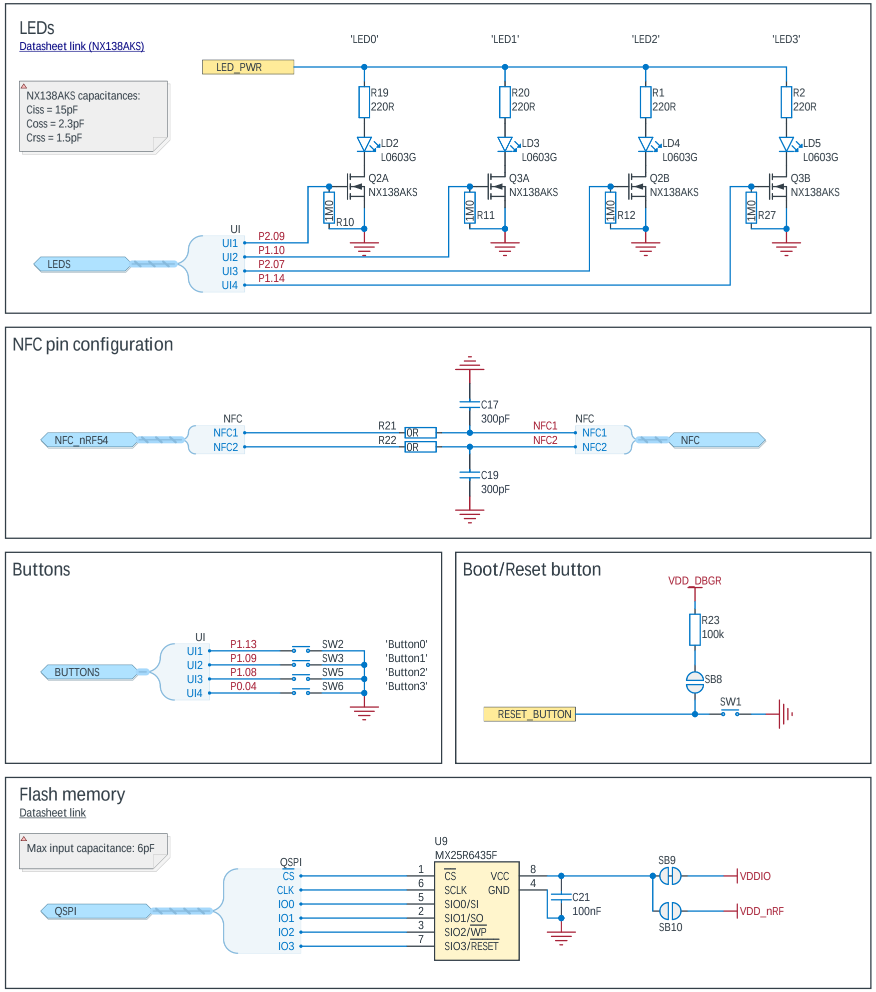
Flow control and Interface control signal controlled by IMCU

Level Shifters and switches

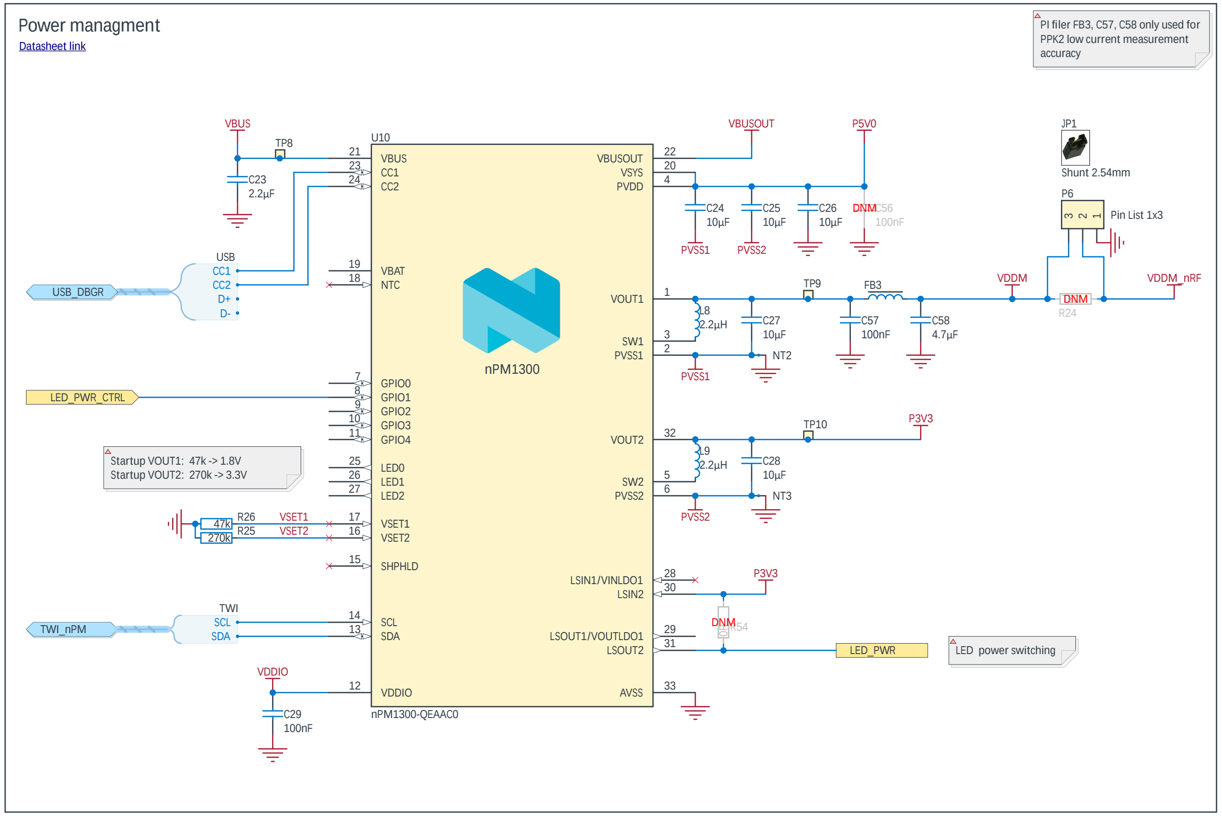
Datasheet link (FSA2466UMX)



Title	nRF54L15-DK - Level Shifters	
Size: A3	PCB Assembly Number: PCA10156	Revision: 0.9.2
Date: 02.10.2024		Sheet 4 of 7
File: pca10156_sheet4_level_shifters.SchDoc		Drawn By: STL1
Classification: PUBLIC		



Title		
nRF54L15-DK - Misc		
Size: A3	PCB Assembly Number: PCA10156	Revision: 0.9.2
Date: 02.10.2024		Sheet 5 of 7
File: pca10156_sheet5_misc.SchDoc		Drawn By: STL1
Classification: PUBLIC		

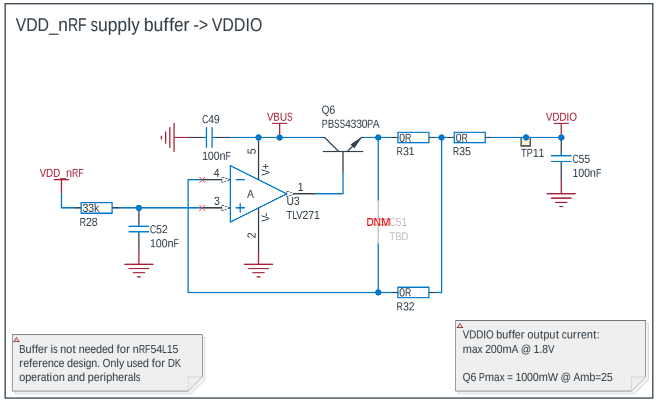


Power domain overview:

VDDM_nRF	Main input power for nRF54L15
VDD_nRF	Main IO VDD for nRF54L15, 1.8V to 2.6V
VDDM	1.8V - 3.3V VDDM source from nPM1300 VOUT1
VDDIO	Buffered VDD_nRF used for IO supply
VDD_DBGR	Debugger VDD tapped from VDDIO
VBUS	USB VBUS after power switch
VBUSOUT	VBUSOUT from nPM1300, used by Debugger
P5V0	5V VSYS (VBUS) used for input to Buck regulators
P3V3	Regulated 3.3V used for analog switches and LEDs

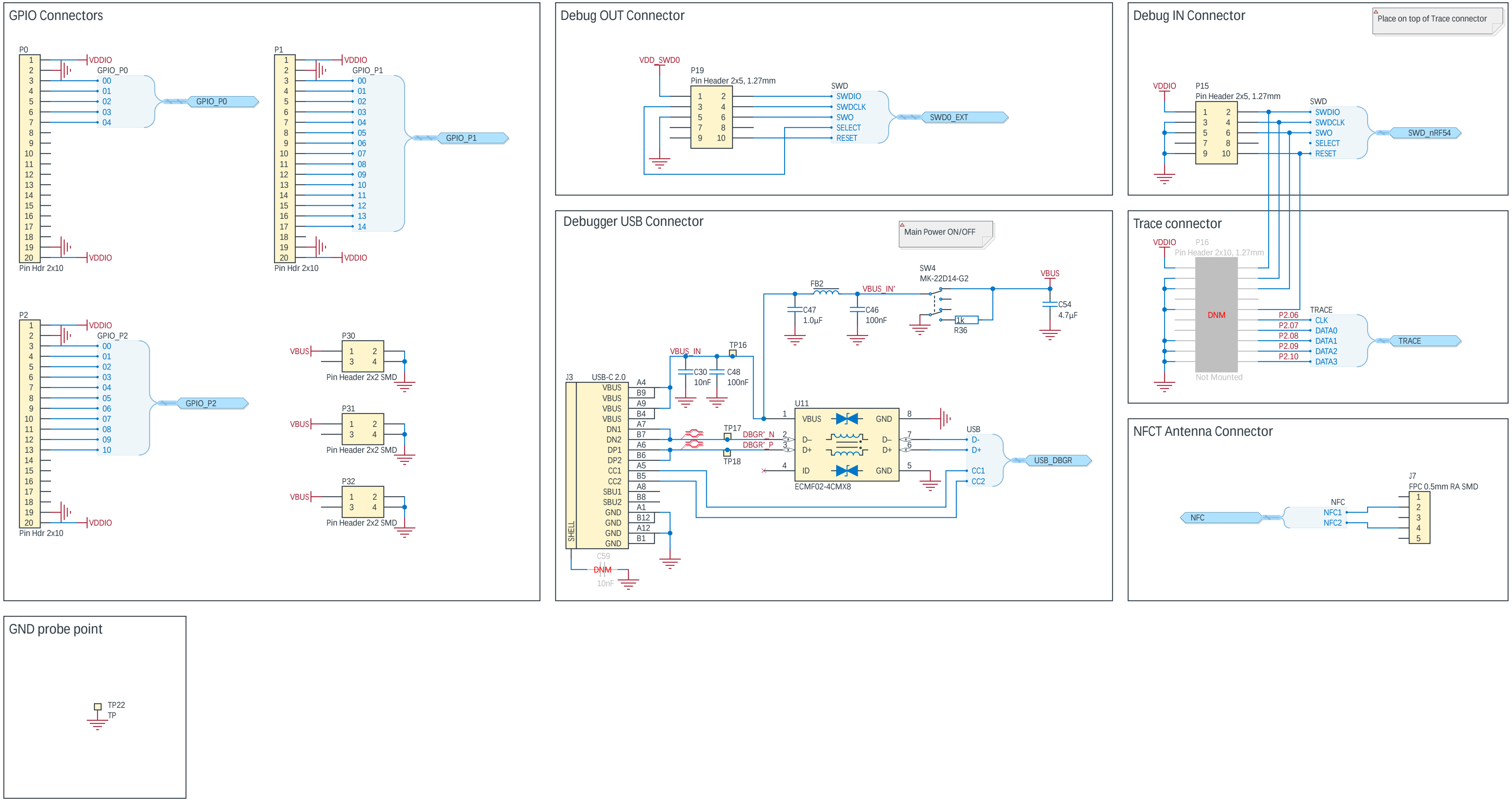
nPM1300 settings:

- VOUT1 set to 1.8V
- VOUT2 set to 3.3V
- VOUT1 and VOUT2 enabled and forced in PWM mode
- VOUT1 and VOUT2 output discharge enable
- Enable LS2 for LED power switching, using GPIO1 as enable/disable signal.



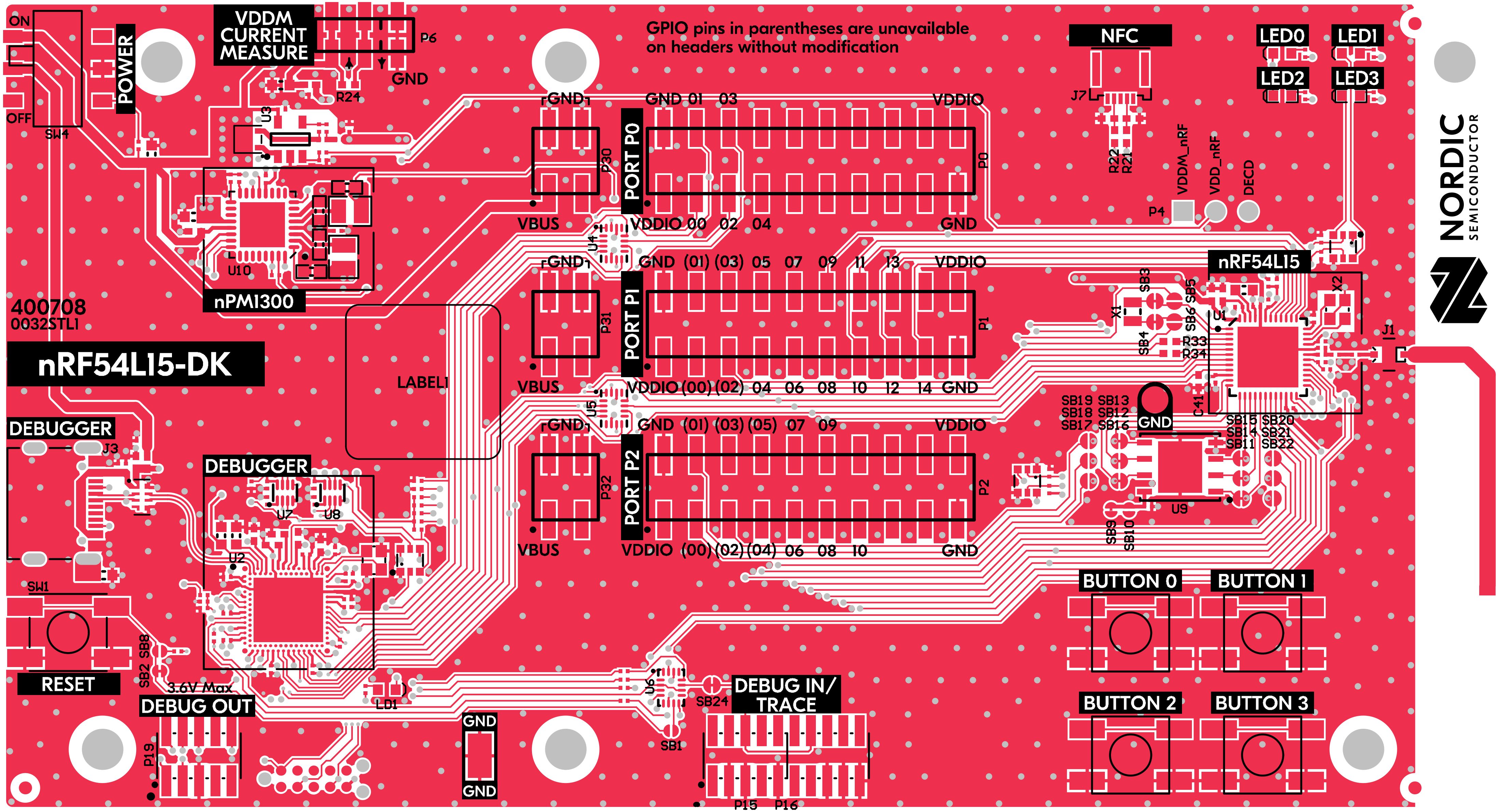
Title: nRF54L15-DK - Power Supply		
Size: A3	PCB Assembly Number: PCA10156	Revision: 0.9.2
Date: 02.10.2024		Sheet 6 of 7
File: pca10156_sheet6_power_supply.SchDoc		Drawn By: STL1
Classification: PUBLIC		

NORDIC SEMICONDUCTOR

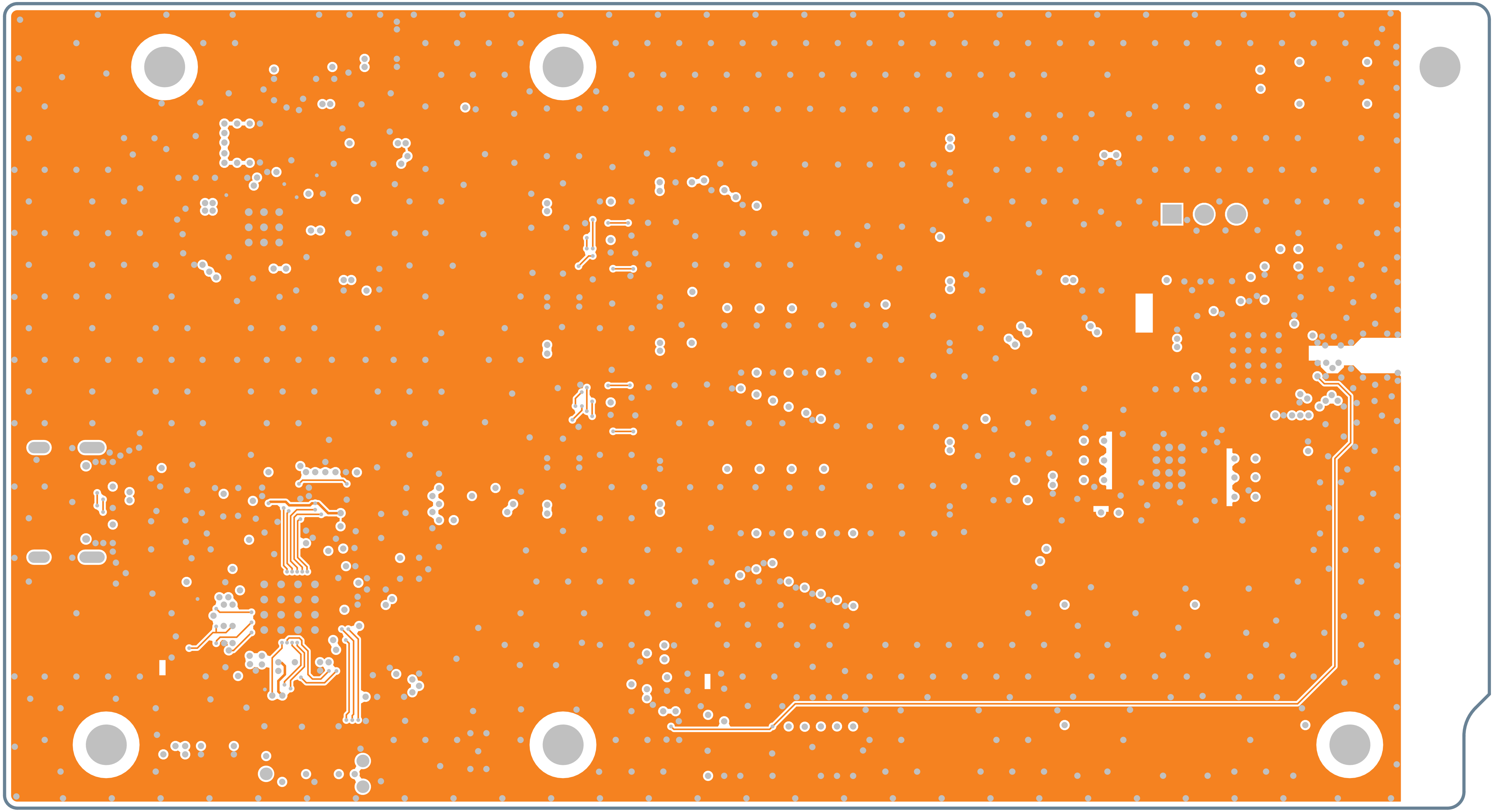


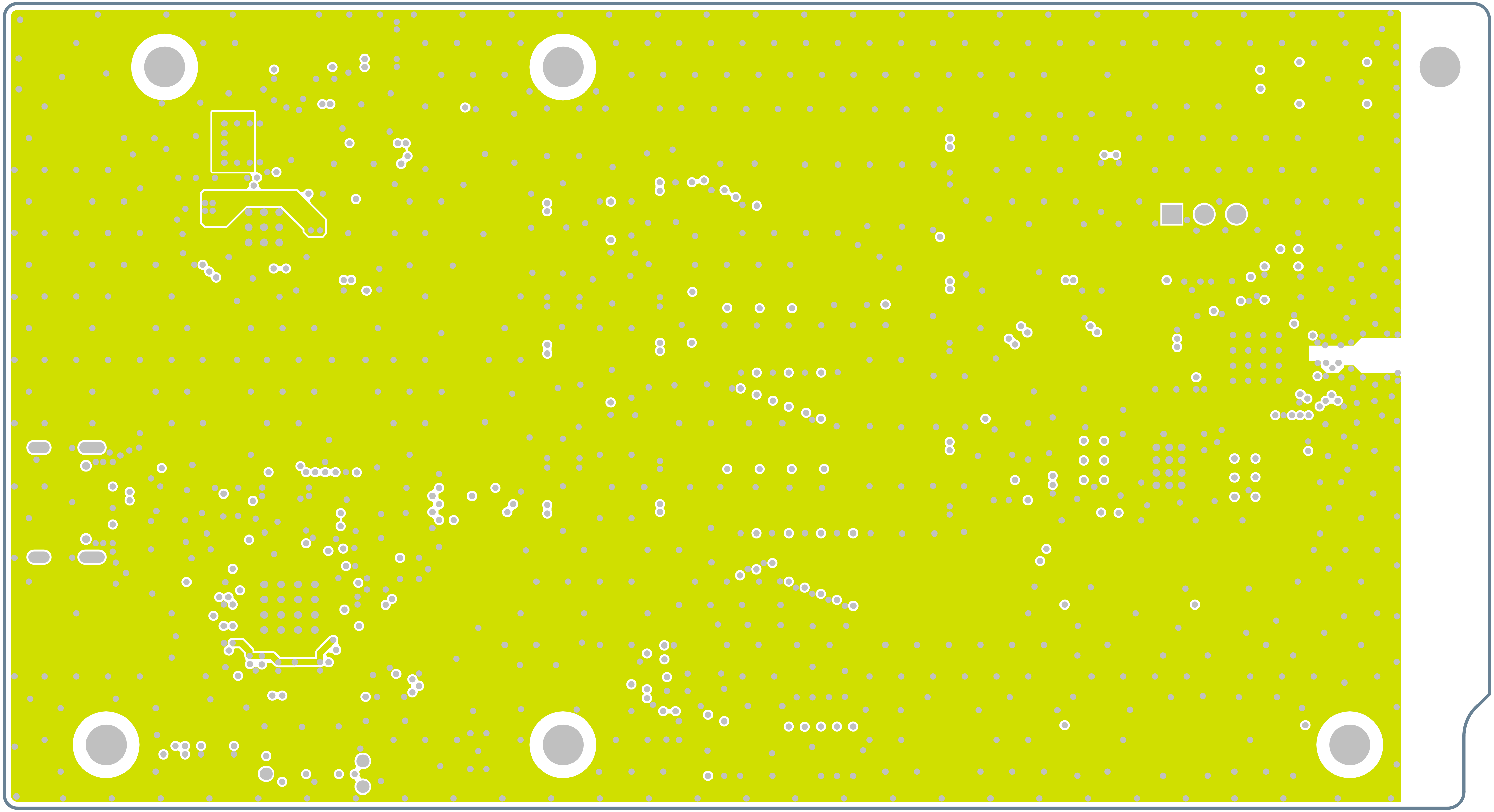
Title: nRF54L15-DK - Connectors		
Size: A3	PCB Assembly Number: PCA10156	Revision: 0.9.2
Date: 02.10.2024		Sheet 7 of 7
File: pca10156_sheet7_connectors.SchDoc		Drawn By: STL1
Classification: PUBLIC		

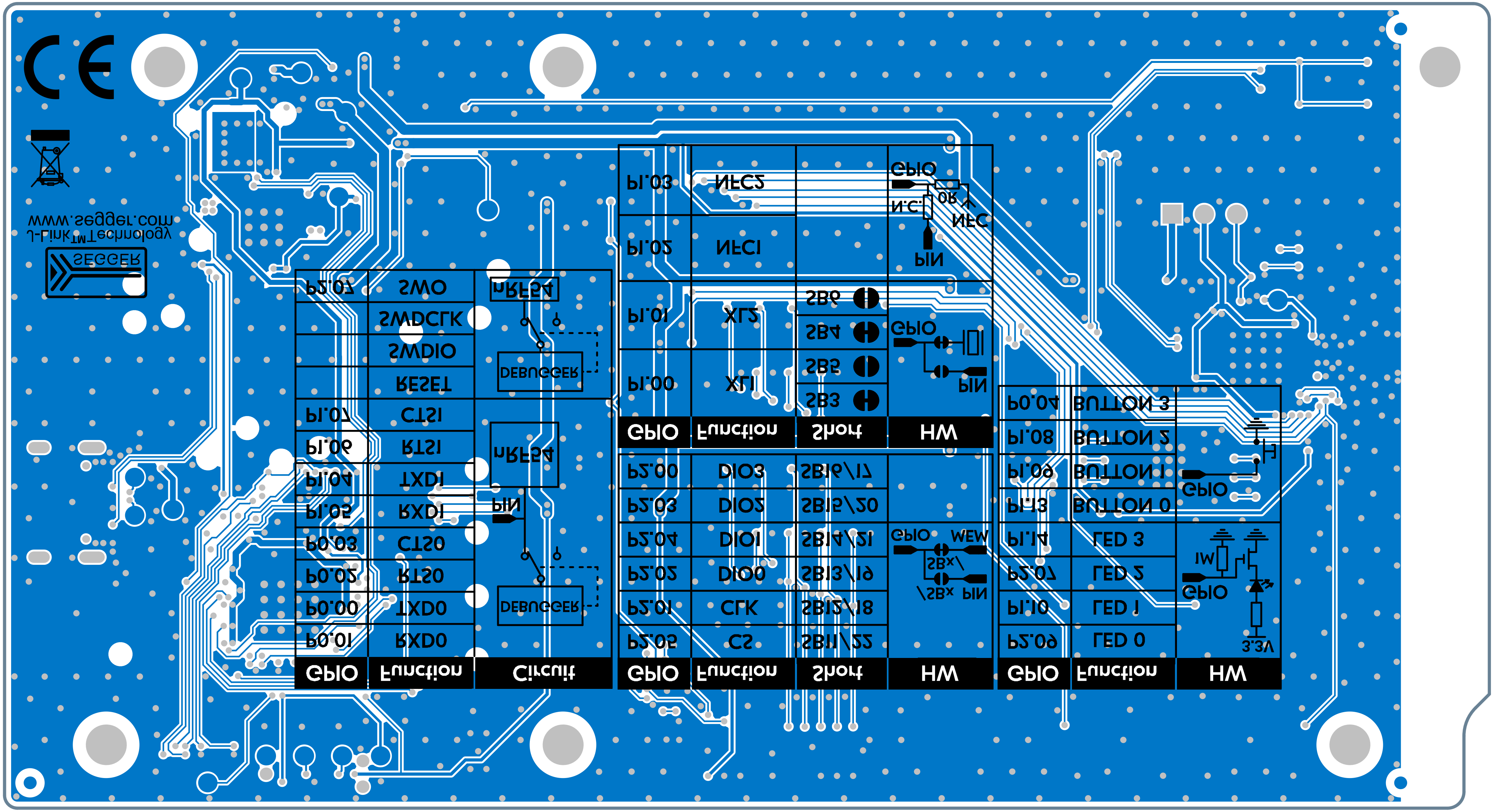
NORDIC SEMICONDUCTOR

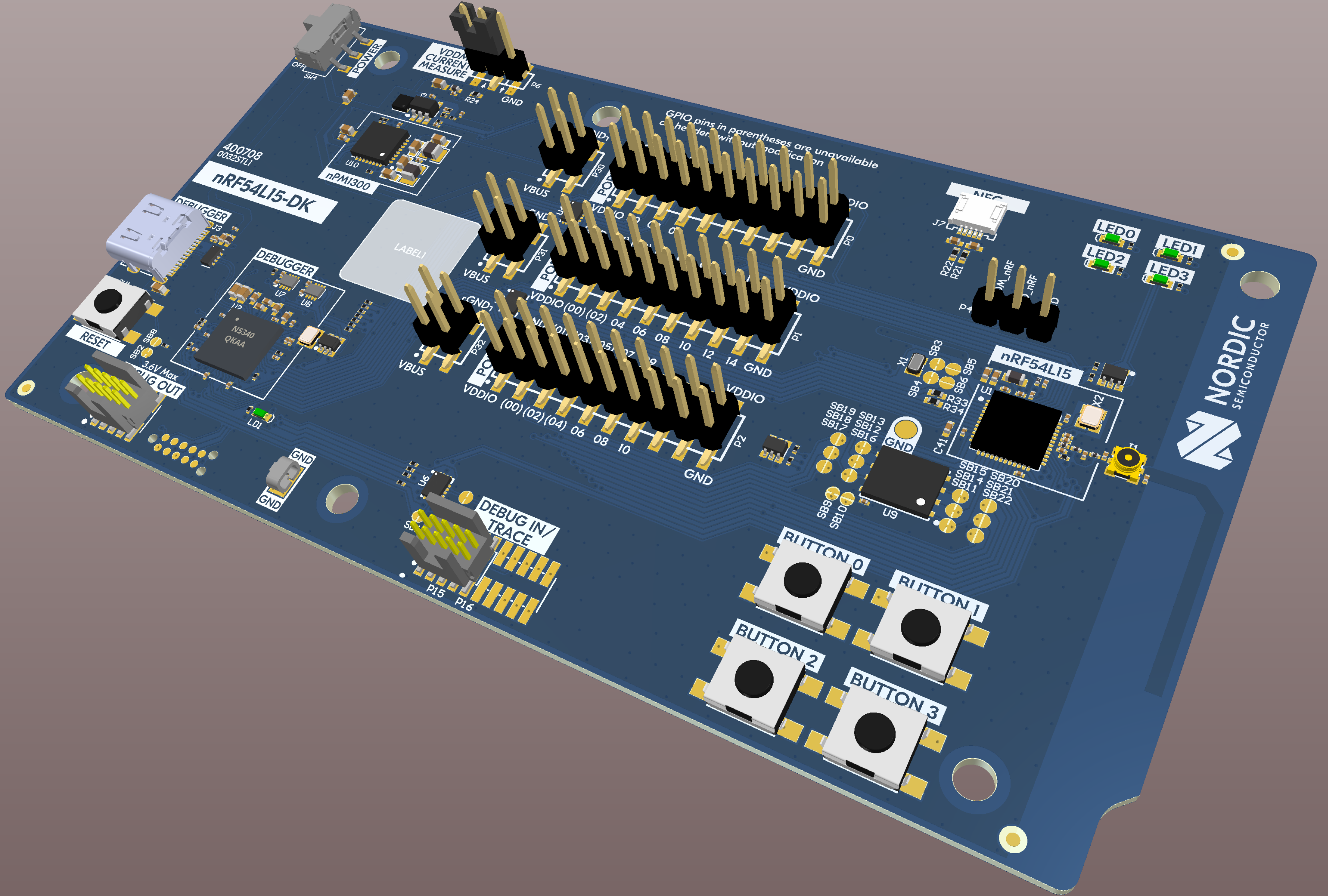


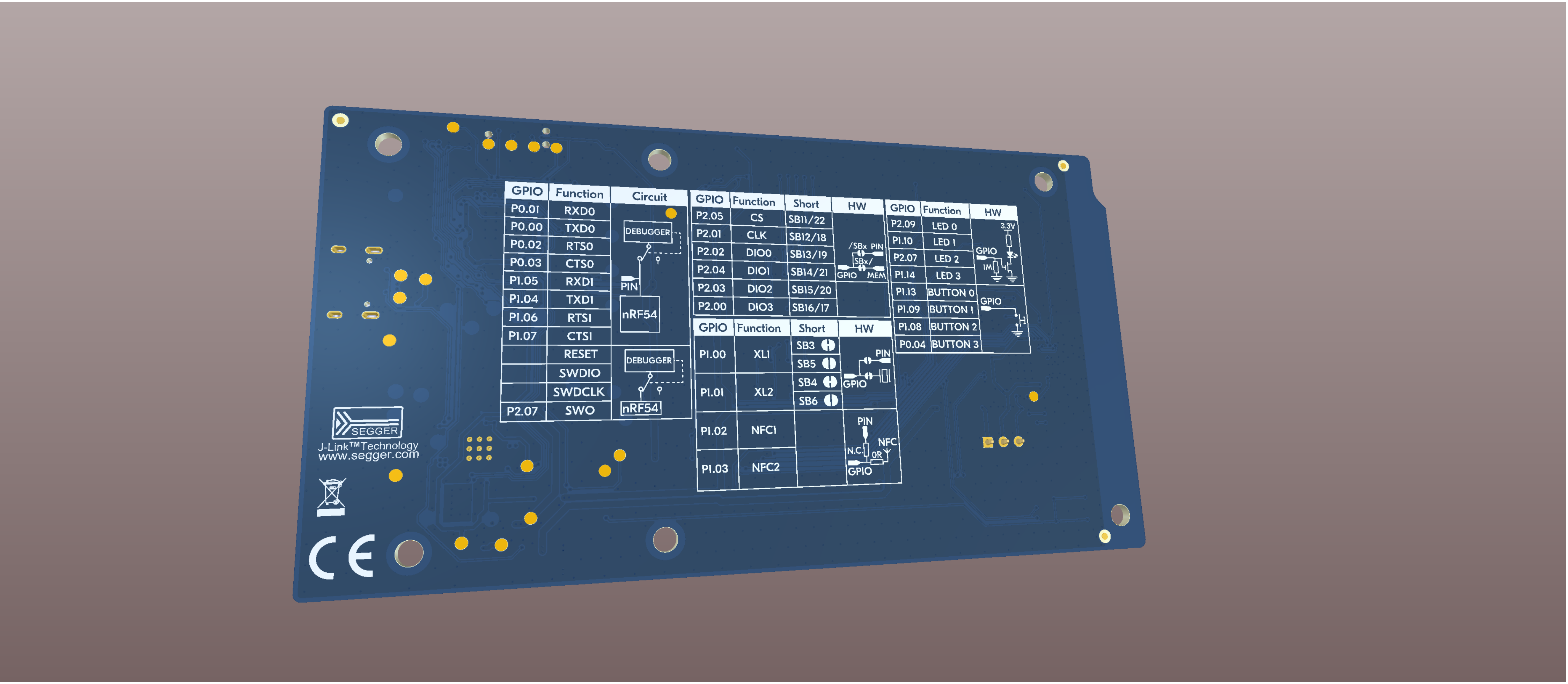
nRF54L15-DK
400708
0032STL1
GPIO pins in parentheses are unavailable on headers without modification
VDDM CURRENT MEASURE
POWER
ON
OFF
SW4
P6
GND
R24
U3
U10
nPM1300
NFC
J7
R22
R21
LED0
LED1
LED2
LED3
VDDM_nRF
VDD_nRF
DECD
P4
NORDIC SEMICONDUCTOR
PORT P0
PORT P1
PORT P2
P0
P1
P2
P30
P31
P32
GND 01 03 VDDIO
VDDIO 00 02 04 GND
GND (01) (03) 05 07 09 11 13 VDDIO
VDDIO (00) (02) 04 06 08 10 12 14 GND
GND (01) (03) (05) 07 09 VDDIO
VDDIO (00) (02) (04) 06 08 10 GND
VBUS
U4
U5
LABEL1
nRF54L15
U1
X1
X2
J1
C41
SB3
SB5
SB6
SB4
R33
R34
SB19 SB13
SB18 SB12
SB17 SB16
SB15 SB20
SB14 SB21
SB11 SB22
U9
SB9
SB10
DEBUGGER
J3
U7
U8
U2
SW1
RESET
SB8
SB2
3.6V Max
DEBUG OUT
P19
LD1
U6
SB24
SB1
DEBUG IN/ TRACE
P15
P16
BUTTON 0
BUTTON 1
BUTTON 2
BUTTON 3






CE
www.segger.com
J-Link™ Technology
SEGGER
GPIO Function Circuit
P2.07 SWO
SWDCLK
SWDIO
RESET
P1.07 CTS1
P1.06 RTS1
P1.04 TXD1
P1.05 RXD1
P0.03 CTS0
P0.02 RTS0
P0.00 TXD0
P0.01 RXD0
nRF54
DEBUGGER
PIN
GPIO Function Short HW
P1.03 NFC2
P1.02 NFC1
P1.01 XL2 SB6
SB4
P1.00 XL1 SB5
SB3
GPIO
0R
N.C.
NFC
PIN
P2.00 DIO3 SB16/17
P2.03 DIO2 SB15/20
P2.04 DIO1 SB14/21
P2.02 DIO0 SB13/19
P2.01 CLK SB12/18
P2.05 CS SB11/22
MEM
SBx/
/SBx
GPIO Function HW
P0.04 BUTTON 3
P1.08 BUTTON 2
P1.09 BUTTON 1
P1.13 BUTTON 0
P1.14 LED 3
P2.07 LED 2
P1.10 LED 1
P2.09 LED 0
1M
3.3V


nRF54L15-DK
400708
DEBUGGER
nPM1300
POWER
RESET
DEBUG IN/
TRACE
VBUS
VDDIO
GND
LED0
LED1
LED2
LED3
nRF54L15
BUTTON 0
BUTTON 1
BUTTON 2
BUTTON 3
NORDIC
SEMICONDUCTOR


GPIO
Function
Circuit
P0.01 RXD0
P0.00 TXD0
P0.02 RTS0
P0.03 CTS0
P1.05 RXD1
P1.04 TXD1
P1.06 RTS1
P1.07 CTS1
RESET
SWDIO
SWDCLK
P2.07 SWO
DEBUGGER
PIN
nRF54
DEBUGGER
nRF54
GPIO Function Short HW
P2.05 CS SB11/22
P2.01 CLK SB12/18
P2.02 DIO0 SB13/19
P2.04 DIO1 SB14/21
P2.03 DIO2 SB15/20
P2.00 DIO3 SB16/17
/SBx PIN
SBx/
GPIO MEM
GPIO Function Short HW
P1.00 XL1 SB3 SB5
P1.01 XL2 SB4 SB6
P1.02 NFC1
P1.03 NFC2
PIN
GPIO
PIN
NFC
N.C.
0R
GPIO
GPIO Function HW
P2.09 LED 0
P1.10 LED 1
P2.07 LED 2
P1.14 LED 3
P1.13 BUTTON 0
P1.09 BUTTON 1
P1.08 BUTTON 2
P0.04 BUTTON 3
3.3V
GPIO
1M
GPIO
SEGGER
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CE